In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

# JEE(Main)-2017

## ANSWERS, HINTS & SOLUTIONS

### FULL TEST – I
### (Main)

| S. No. | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | **C** | D | A |
| 2. | **B** | D | B |
| 3. | A | B | B |
| 4. | C | D | B |
| 5. | **C** | D | B |
| 6. | **D** | C | A |
| 7. | D | C | **B** |
| 8. | A | D | B |
| 9. | C | A | A |
| 10. | B | B | B |
| 11. | A | C | C |
| 12. | **C** | B | A |
| 13. | A | D | A |
| 14. | **D** | D | D |
| 15. | C | D | A |
| 16. | A | A | A |
| 17. | C | C | C |
| 18. | D | D | C |
| 19. | A | D | **C** |
| 20. | B | D | D |
| 21. | **D** | B | D |
| 22. | A | B | D |
| 23. | **B** | B | D |
| 24. | D | D | B |
| 25. | A | C | D |
| 26. | B | D | A |
| 27. | B | D | B |
| 28. | **A** | C | D |
| 29. | D | B | B |
| 30. | B | **C** | A |

# Physics

## PART – I

### SECTION – A

1. The time taken in first a/2 is T/6 and in next a/2 will be T/4. So

$$t = \frac{T}{6} + \frac{T}{4} = \frac{5}{12}T$$

$$= \frac{5}{12} \times 2\pi\sqrt{\frac{M}{k}} = \frac{5\pi}{6}\sqrt{\frac{m}{k}}$$

2. MI about CM = $mR^2 + 2\,mR^2/4$
$I = 3/2\ mR^2$
Conservation of AM about CM,
$Mv_o R/4 = 3/2\ mR^2\omega \quad \Rightarrow \omega = v_o/6R$

3. $t = \frac{10 + x}{C} = \frac{4 - x}{C}$

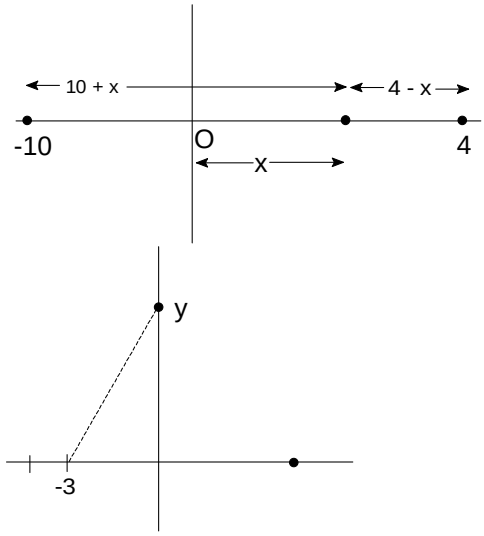
$\therefore -6 = 2x$

$t = \frac{10 - 3}{C} \qquad \therefore x = -3$

$\therefore t = \frac{7}{C}$

$\therefore 7 = \sqrt{3^2 + y^2}$

$\therefore y = \sqrt{40}$




4. We know gravitational potential energy is

$$U = -\frac{3}{5}\frac{GM^2}{R}$$

Volume of star $V = \frac{4\pi R^3}{3} \qquad \therefore R = \left(\frac{3V}{4\pi}\right)^{\frac{1}{3}}$

$$\therefore U = -\frac{3}{5}\left(\frac{4\pi}{3V}\right)^{\frac{1}{3}} GM^2$$

Pressure $P = -\frac{\partial u}{\partial V}$

$$= -\frac{1}{5}\left(\frac{4\pi}{3}\right)\frac{GM^2}{V^{\frac{4}{3}}}$$

$$\therefore P \propto V^{-\frac{4}{3}}$$

5. $V_a - 3 + \frac{18}{3+R} = V_b$

$V_a - V_b = 3 - \frac{18}{3+R} = 0$

$R = 3\Omega$

6. $\Delta u$ in capacitor = 0

Work by battery = qE = 64 $\mu$J

$\therefore$ Heat = 64 $\mu$J

7. Time taken for the plate to fall is $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 40}{980}}$

$\therefore t = \frac{2}{7}\text{sec}$

time period $T = \frac{t}{8} = \frac{2}{8\times 7} = \frac{1}{28}\text{sec}$

$\therefore$ frequency $f = \frac{1}{T} = 28$ Hz

8. For the water level to remain stationary

Volume efflux = rate of filling (x)

$\therefore VA = x$

$\sqrt{2gh}\ A = x \qquad \therefore h = \frac{x^2}{2gA^2}$

$h = \frac{(70)^2}{2\times 980\times (0.25)^2}$

$h = 40$ cm

9. (i), (ii) are true but (iii) is false, as we know that viscosity in gaseous is about 100 times less than viscosity in liquids.

10. Shape of the composite bar is convex toward PQ if $\alpha_1 < \alpha_2$.

11. Voltage gain = $\frac{V_{out}}{V_{in}} = \frac{I_{out}}{I_{in}} + \frac{R_{out}}{R_{in}}$

$= \frac{2\times 10^{-3}}{40\times 10^{-6}} \times \frac{4\times 10^3}{100}$

$= 2000$

12. When $\frac{dE}{dt} = 0$

E is maximum

13. least count of Vernier caliper

14. Force on the upper block = *ma*

Work done = F.S = *ma*

15. Collector current $I_c$ =

$\frac{\text{voltage drop across} R_L}{R_L} = \frac{1}{1000} = 10^{-3}$ amp

Now $V_{CE} = 9 - 1 = 8$ volt

Current gain $\beta = \frac{I_C}{I_B}$

or $\frac{25}{26} = \frac{10^{-3}}{I_B}$ $\quad \therefore I_B = 1.04 \times 10^{-3}$ amp.

Voltage gain $= \beta \frac{R_L}{R_C} = \frac{25}{26} \times \frac{1000}{200} = \frac{125}{26}$

Power gain $= \beta^2 \frac{R_L}{R_C} = \left(\frac{25}{26}\right)^2 \times \frac{1000}{200} = 4.6$

Again $I_E = I_B + I_C = 1.04 \times 10^{-3} + 10^{-3} = 2.04 \times 10^{-3}$ A

16. Magnification $M = \frac{F}{f} = 9 \quad \therefore F = 9f \quad .....(i)$

Distance between lenses d = F + f $\quad \therefore F + f = 20 \quad .....(ii)$

From equation $(i)$ & $(ii)$ we get F = 18 cm

f = 2 cm

17. The objective has a shorter focal length. The objective has a shorter aperture.

18. Outgoing volume /sec = incoming volume per sec

$Q + 3 \times 10^{-6} + 4 \times 10^{-6} + 1 \times 10^{-6} = 4 \times 10^{-6} + 8 \times 10^{-6} + 2 \times 10^{-6} + 7 \times 10^{-6}$

$Q = 13 \times 10^{-6} m^3 / s$

19. Here the width of principal maxima is 2.5 mm

Then its half width is = $\frac{\beta}{2}$

$= \frac{2.5}{2} = 1.25 \times 10^{-3} m$

Diffraction angle $\theta = \frac{\left(\frac{\beta}{2}\right)}{D} = \frac{1.25 \times 10^{-3}}{2} \quad .....(i)$

$\therefore a\theta = \lambda \quad \therefore \theta = \frac{\lambda}{a} \quad .....(ii)$

from $(i)$ & $(ii)$

$\lambda = \theta a$

$= \frac{1.25 \times 10^{-3}}{2} \times 10^{-3}$

$= 6.25 \times 10^{-7} m$

$\therefore \lambda = 6250 \overset{0}{A}$

20. Average energy density is given by,

$$U = \frac{1}{2}\in_0 E^2 = \frac{1}{2}\in_0 \left(\frac{E_0}{\sqrt{2}}\right)^2 = \frac{1}{4}\in E^2$$

$$U = \frac{1}{4}\times 8.85\times 10^{-12}\times 1^2$$

$$U = 2.2\times 10^{-12}\,J/m^3$$

21. $\vec{\tau} = \hat{i}\times B\,\hat{j} = B\hat{k}$

22. For strong reflection, the least optical path difference introduced by the film should be $\frac{\lambda}{2}$. The optical path difference between the waves reflected from the surfaces of the film is $2\mu d$.

Thus for strong reflection $2\mu d = \frac{\lambda}{2}$

$$\therefore d = \frac{\lambda}{4\mu} = \frac{589}{4\times 1.25} = 118\text{ nm}$$

23. $$\frac{E_0}{\sqrt{2}} = \sqrt{V_1^2 + (V_2 - V_1)^2} = 50V$$

$$\therefore E_0 = 50\sqrt{2}\,V$$

24. $$y = x_0\cos 2\pi\left(nt - \frac{x}{\lambda}\right)$$

$$y = x_0\cos\frac{2\pi}{\lambda}(Vt - x) \qquad \because V = n\lambda$$

$$\left(\frac{dy}{dt}\right)_{max} = x_0\frac{2\pi}{\lambda}(v) = 2V \quad (\text{given})$$

$$\therefore \lambda = \pi x_0$$

25. $$M = -\frac{V_0}{U_0}\frac{D}{fe}$$

here $u_0 = -\left(\frac{54}{131}\right)$ cm

$$M = -\frac{13.5}{\left(\frac{54}{131}\right)}\times\frac{25}{2.5}$$

$$= -327.5$$

26. Energy $\propto (\text{Amplitude})^2$

$E = KA^2$, $\quad K\,(\text{constant})$

$$\frac{dE}{E} = 2\frac{dA}{A} = \frac{2\times 5}{100} = 10\%$$

27. we know $b = \frac{r}{2m} = \frac{1.5}{2\times 25\times 10^{-3}} = 30$

& $\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{25}{25\times10^{-3}}} = \sqrt{1000}$

$\therefore$ time period

$$T = \frac{2\pi}{\sqrt{\omega^2 - b^2}} = \frac{2\pi}{\sqrt{1000-900}} = 0.628 \text{ sec}$$

28. Net Power given to N2 gas = 100 – 30 = 70 cal/sec
The nitrogen gas expends isobarically.

$\therefore \quad Q = nC_p\frac{dT}{dt}$ or $70 = 5\times\frac{7}{2}R\frac{dT}{dt}s$

$\therefore \quad \frac{dT}{dt} = 2K/\text{sec}$

29. If I is the final intensity and $I_0$ is the initial intensity then

$I = \frac{I_0}{2}\left(\cos^2 30\right)^5$ or $\frac{I}{I_0} = \frac{1}{2}\times\left(\frac{\sqrt{3}}{2}\right)^{10} = 0.12$

$\therefore$ % change $= 12\%$

30. Audible waves.

# Chemistry

## PART – II

### SECTION – A

1. Normality of $H_2O_2$ = 11.2/5.6

$\therefore \frac{11.2}{5.6} \times V = \underset{\text{moles of } Fe^{+2}}{4} \times 1$

Normality of $H_2O_2$

V = 2 L

2. $n_1 = 4 \quad \because 4 \to 2$ produces 3 different lines.

$n_1 = 4 \Rightarrow \ell = 0, 1, 2, 3$

$\therefore$ orbital angular momentum $= \sqrt{\ell(\ell+1)}\,\hbar$

$= 0$

or $\sqrt{2}\,\hbar$ or $\sqrt{6}\,\hbar$ or $\sqrt{12}\,\hbar$

3. $\frac{r_1}{r_2} = \frac{K[P]^n}{K(8[P])^n}$

$2 = 8^n$

$n = \frac{1}{3}$

4. $\Delta U = \Delta H - \Delta nRT$

=41000 - $1 \times 8.314 \times 373$

$= 37.9 \text{ kJ mol}^{-1}$

5. $U_{mps}(A) < U_{mps}(B)$

$U_{mps} \propto \frac{1}{\sqrt{M}}$

$\therefore M_A > M_B$

$\because r \propto \frac{1}{\sqrt{M}}$

$\therefore r_A > r_B$

6. No. of A atoms = $\left(\frac{1}{8}\right) \times 7 = \frac{7}{8}$

No. of B atoms = $6 \times \frac{1}{2} = 3$

$\therefore$ A :B $= \frac{7}{8} : 3 = 7 : 24$

7. $H_2 \to 2H^+ + 2e^-$

$$E_{ox} = E^{o}_{ox} - \frac{0.0591}{2}\log\left[H^{+}\right]^{2}$$

$pH = 4.23$

8. T ↑ Adsorption ↓

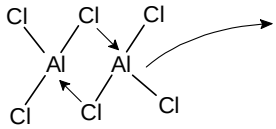
9. $\pi V = nST$

$500\ mm \times V_1 = nS \times 283$

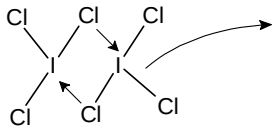
$105.3\ mm \times V_2 = nS \times 298$

$$\frac{V_2}{V_1} = \left(\frac{298}{105.3} \times \frac{500}{283}\right)$$

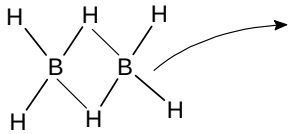
$= 5$

10. (a) 

→ $sp^3$ non - planar

(b) 

→ $sp^3d^2$ hybridization here square planar

(c) 

→ $sp^3$ (non - planar)

(d) tub – like structure (non – planar)

11. (i) exothermic: (i), (v) & (vi)
(ii) endothermic: (ii), (iii) & (iv)

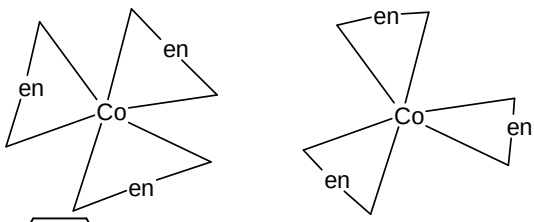
12. $M^{+}\left(NH_3\right) + e^{-}\left(NH_3\right) + NH_3\left(\ell\right) \longrightarrow MNH_2 + H_2\left(g\right)$

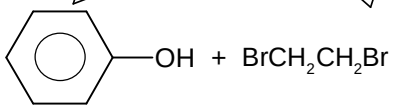
13. comparing D – O & H – O bond strength

14. $E_a$ of 1st step must be higher than 2nd step and XY has lower energy than X + Y.

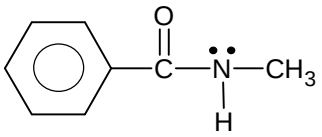
15. $I^{-} \longrightarrow I_2\ \left(\text{oxidized}\right)\ \&\ S^{+6} \longrightarrow S^{+4}\left(\text{Reduced}\right)$

16. No catalyst is used in thermite process

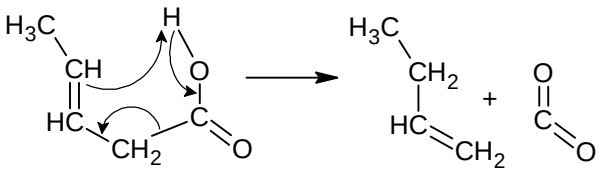
17. 


18. 

19. In Hoffmann Bromamide reaction N-alkyl substituted amide.



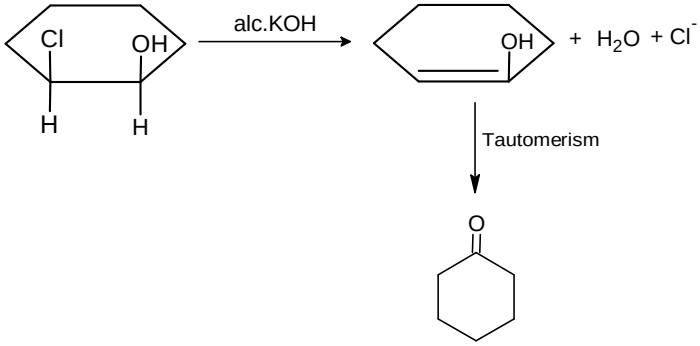

→ will not be replaced so it dosent give hoffman bromamide reaction.

20. 


6-membered Transition state

21. $2\alpha - H$ are required for driving the equilibrium forward & getting enolate of an ester.

22. Esterification reaction

23. 


24. all will produce racemic mixture, $\because$ all of them have planar intermediates.

25. II$^{nd}$ is non superimposable mirror image of (I).

26. A, B; planar molecule.
D-Allene, non planar with no plane/ point of symmetry.

27. octet complete.

28. It is converted into an aldehyde.

29. $K_p = \dfrac{\left(X_{Cl_2}\right)\left(X_{PCl_3}\right)}{\left(X_{PCl_5}\right)} P_T$

30. let solubility of $A_2B_3$ be x M $\therefore \left[A^{+3}\right] = 2x = S \therefore x = \dfrac{S}{2}M$

# Mathematics

## PART – III

### SECTION – A

1 $x = r\sec\theta \qquad y = r\tan\theta \qquad \left(r^2 = x^2 - y^2\right)$

$$\frac{rdr}{r^2\sec\theta d\theta} = r\sec\theta$$

$$\int\frac{dr}{r^2} = \int\sec^2\theta d\theta$$

$$-\frac{1}{r} = \tan\theta + c$$

$$-\frac{1}{\sqrt{x^2-y^2}} = \frac{y}{\sqrt{x^2-y^2}} + c$$

Put $y = 0$ at $x = 1$, $\quad C = -1$

so $\quad y + 1 = \sqrt{x^2 - y^2}$

2. $$\frac{\left(1+\frac{\sqrt{x}}{2}\right)^{40} + \left(1-\frac{\sqrt{x}}{2}\right)^{40}}{2} = \left[nC_0 + nC_2\left(\frac{\sqrt{x}}{2}\right)^2 + nC_4\left(\frac{\sqrt{x}}{2}\right)^4 + .....\right]$$

Put x = 1,

$$\frac{\left(1+\frac{1}{2}\right)^{40} + \left(1-\frac{1}{2}\right)^{40}}{2} = \frac{\frac{3^{40}}{2^{40}} + \frac{1}{2^{40}}}{2}$$

$$= \frac{3^{40}+1}{2^{41}}$$

3. $\left(2x + y - 6\right) + \lambda\left(2y - 3x - 12\right) = 0$

so fixed point where $\ell_1$ & $\ell_2$ will pass is,

$2x + y - 6 = 0$

$2y - 3x - 12 = 0$

we get, $(0, 6)$

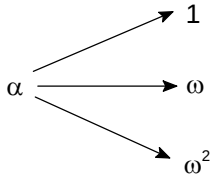
slope of line $\ell_1 = -\left(\frac{1}{\frac{3}{-4}}\right) = \frac{4}{3}$

slope of line $\ell_2 = -\left(\frac{r}{s}\right) = -\frac{1}{5}$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \frac{23}{11} \Rightarrow \theta = \tan^{-1}\left(\frac{23}{11}\right)$$

4. $$\frac{p}{q}\times\frac{q}{r}\times\frac{r}{p} = 1$$

$\Rightarrow \alpha^3 = 1 \Rightarrow$




(not possible b/c distinct p,q,r)

$\Rightarrow \alpha = \omega$ & $\alpha = \omega^2$

$\Rightarrow \omega + \omega^2 = -1$

5. $f'(x) = \dfrac{f(x+h) - f(x)}{h}$

$= \dfrac{f(h) + (x^2 + h^2)xh}{h}$

$f'(x) = \lim\limits_{h \to 0} \dfrac{f(h)}{h} + x(x^2 + h^2)$

$= 2 + x^3$

$f(x) = 2x + \dfrac{x^4}{4} + K$

for K, $\lim\limits_{n \to u} \dfrac{f(x)}{x} = 2$

$\dfrac{2x + \frac{x^4}{4} + K}{x} = 2 \Rightarrow k = 0$

so, $f(x) = 2x + \dfrac{x^4}{4}$

6. $S = \{x, y : 0 \le x, y \le 60\}$

$A \rightarrow$ Event that $x - y \le 10$

$B \rightarrow$ Event that $y - x \le 10$

$n(E) = 60^2 - 2 \times \dfrac{1}{2} \times 50 \times 50$

$P(E) = \dfrac{1100}{3600}$

7. $n(E) = \dfrac{1}{2}\left[{}^4C_1 . {}^2C_1 + {}^{24}C_1 . {}^3C_1 + {}^{36}C_1 . {}^4C_1\right] = 112$

$\Rightarrow P(E) = \dfrac{112}{{}^{64}C_2} = \dfrac{1}{18}$

8. Divide by $2^x$,

$2^x + 2^{-x} = 5\cos x - 3$

$\underset{(\ge 0)}{2^x + 2^{-x}} - 2 = \underset{(\le 0)}{5(\cos x - 1)}$

$\cos x - 1 = 0$ & $2^x + 2^{-x} = 2$

$\Rightarrow x = 0, 2\pi, 4\pi$ & $x = 0$

Inter sec tion of both solution we get $x = 0$

Number of solution $= 1$

9. $\dfrac{f'(x)}{f(x)} = \sum_{n=1}^{50}, n(51-n)\left(\dfrac{1}{x-n}\right)$

$\dfrac{f'(51)}{f(51)} = \sum_{n=1}^{50} n(51-n) \times \dfrac{1}{(51-n)}$

$\dfrac{f(51)}{f'(51)} = \dfrac{1}{1275}$

10. $f'(n) = e^{2n^3} + 3n^2 + 6x - 5\left(6x^2 + 6x + 6\right)$

$= 6\left(x^2 + x + 1\right)e^{2x^3+3x^2+6x-5} > 0$

and $e^{2x^3+3x^2+6x-5}$ will not be negative so one - one into

11. $\int \dfrac{\cos ec^2 x - n}{\cos^n x} dx = \int \left(\cos ec^2 x - n\right)\sec^n x \, dx$

$= \int \left(\cos ec^2 x\right)\sec^n x \, dx - \int n \sec^n x \, dx$

using by part

we get $= -\sec^{(n-1)} x \cos ec \, x$

so, $f(x) = \cos ecx$ & $9(x) = \sec x$

and min value of $\sqrt{\sec^2 x + \cos ec^2 x} = 4$

so $(4)^n = (2)^{2n}$

12. Let $f(x) = \sum_{r=-1}^{m}\left(x - \dfrac{1}{2^r}\right)^{2n+1}$

After differentiation f'(x) is positive so f(x) is increasing so only one real root & other are imaginary.

13. sum of division $= \underset{\downarrow}{(2)} \qquad \left(3^0 + 3^1 + 3^2 + 3^3\right)\left(5^0 + 5^1 + 5^2\right)\left(7^0 + 7^1\right)$

exact 2 will come

$= 2 \times 40 \times 31 \times 8$

$= 19840$

14. $(p \wedge \sim q) \Rightarrow r$

converse $r \Rightarrow (p \wedge \sim q)$

inverse $\sim r \Rightarrow \sim (p \wedge \sim q)$

$\sim r \Rightarrow (\sim p \vee q)$

$\sim r \Rightarrow (\sim p \vee q)$

$\sim r = (p \Rightarrow q)$

15. roots are x = 1, 1

Product of root = 1 $\Rightarrow$ $b = \frac{2ac}{a+c}$

$\Rightarrow$ a, b, c are in HP

$\frac{1}{a},\frac{1}{b},\frac{1}{C}$ are in AP

$\frac{S}{a},\frac{S}{b},\frac{S}{C}$ are in AP

$\frac{S-a}{a},\frac{S-b}{b},\frac{S-c}{c}$ are in AP

$\frac{bc}{(S-b)(S-c)},\frac{ca}{(S-c)(S-a)},\frac{ab}{(S-a)(s-b)}$ are in AP

16. This is shifted ellipse $3(x-1)^2+4(x-1)(y+2)+3(y+2)^2=0$

$3X^2+4XY+3Y^2=1$

so rotate $45^0$ we get

$\frac{x^2}{5}-\frac{y^2}{1}=1\Rightarrow A=\frac{\pi}{\sqrt{5}}$

17. Put $e^x = t$

$\int_0^1 t^{121}(1-t)^{19}\,dt$

$I_{121,18}=\frac{1}{(121+19+1)\left({}^{121+19}C_{19}\right)}$

$=\frac{1}{141\times\left({}^{140}C_{19}\right)}=\frac{19!121!}{141!}$

18. 3 vowels can be selected by = 12 ways
Consonant can be selected by 72 ways
Total ways = $6C_3\times12\times72=17280$

19. $E_1\rightarrow n_F=n_B$ and $n_R=n_L+1$

$E_2\rightarrow n_F=n_B$ and $n_L=n_R+1$

$E_3\rightarrow n_R=n_L$ and $n_F=n_B+1$

$E_4\rightarrow n_R=n_L$ and $n_B=n_F+1$

$P(E_1)=P(E_2)=P(E_3)=P(E_4)$

$P(E)=4P(E_1)$

$n(s)=4^9$

Event $E_1$ can occurs in following ways,

| F | B | R | L |
|---|---|---|---|
| 0 | 0 | 5 | 4 |
| 1 | 1 | 4 | 3 |
| 2 | 2 | 3 | 2 |
| 3 | 3 | 2 | 1 |
| 4 | 4 | 1 | 0 |

$n(E_1)={}^9C_5.{}^4C_4+{}^9C_1.{}^8C_1.{}^7C_4.{}^3C_3+{}^9C_2.{}^7C_2.{}^5C_3.{}^2C_2+{}^9C_3.{}^6C_3.{}^3C_2.{}^1C_1+{}^9C_4.{}^5C_4.{}^1C_1=15876$

$P(E)=4\times\left(\frac{15876}{4^9}\right)$

20. $[nx] = (n+1)x$

$nx - \{nx\} = nx + x$

$-x = \{nx\}$

$\Rightarrow x = 0$ (one solution)

since n is the positive integer so Total solution of x is $n+1$

21. $I = \int_0^{\frac{\pi}{8}} \frac{\left(\frac{\pi}{8} - x\right)}{1 + \cos 4\left(\frac{\pi}{8} - x\right) + \sin 4\left(\frac{\pi}{8} - x\right)} dx$

$2I = \frac{\pi}{8}\int_0^{\frac{\pi}{8}} \frac{1}{1 + \cos x + \sin 4x} dx$

$= \frac{\pi}{8}\int_0^{\frac{\pi}{8}} \frac{\sec^2 2x}{2 + 2\tan 2x} dx$

$I = \frac{\pi}{64}\int_0^{\frac{\pi}{4}} \frac{\sec^2 t}{1 + \tan t} dt$

$I = \frac{\pi}{64}\left|\ell n(1 + \tan t)\right|_0^{\frac{\pi}{4}} = \frac{\pi}{64}\ell n2$

$\Rightarrow b = 64$

22. $\int_1^e e^x (x\ell nx + \ell nx) dx$

$\int_1^e e^x (x\ell nx + \ell nx + 1 - 1)\, dx$

$\int_1^e e^x (x\ell nx + (\ell nx + 1)) - \int_1^e e^x dx$

$\left|e^x x\ell nx\right|_1^e - \left|e^x\right|_1^e$

$= e^{e+1} - e^e + e$

$P = e+1, q = e, r = 1$ & so, $p + q + r = e + 1 + e + 1$

$= 2e + 2$

$= 2(e+1)$

23. $a + b > c$ (sum of two side > third side)

$a - c + b > 0 \Rightarrow f(1) > 0$

$f(-2) \geq 0$ also

$4a - 2c + b \geq 0$

$\Rightarrow b \geq 2(-4a)$

24. $x^2-3x+3=1 \quad \& \quad x^2=1$

$x=1,2 \quad ---(i) \quad \& \quad x=\pm 1 \quad -------(ii)$

Taking intersection and we get x = 1.

So only one solution is possible.

25. $A^{-1}=\frac{1}{|A|}\text{adj}(A) \Rightarrow A^{-1}=\begin{bmatrix}1 & -2 & 1\\ 0 & 1 & -2\\ 0 & 0 & 1\end{bmatrix}$

$|A|=1 \quad \& \quad BA=A$

$BAA^{-1}=AA^{-1} \Rightarrow B=I$

Total no of 1 in matrix B = 3 & matrix $A^{-1}=4$

26. $\frac{\sin A}{\sin c}=\frac{\sin(A-B)}{\sin(B-C)}$

$\sin(B+C)\sin(B-C)=\sin(A+B)\sin(A-B)$

$\sin^2 B-\sin^2 C=\sin^2 A-\sin^2 B$

$\Rightarrow \sin^2 A,\sin^2 B,\sin^2 C \Rightarrow AP$

$\sin A\sin A,\sin B\sin B,\sin C\sin C \Rightarrow AP$

$\frac{a}{2R}\sin A,\frac{b}{2R}\sin B,\frac{C}{2R}\sin C \Rightarrow AP$

$a\sin A,b\sin B,c\sin C \Rightarrow AP$

27. let any point on $x^2=4y$ as $(2t,t^2)$ mirror image of $(2t,t^2)$ along line x + y = 2 is (h, k). is

$\frac{h-2t}{1}=\frac{K-t^2}{1}=-2\left(\frac{2t+t^2-2}{2}\right)$

$h=2t-2t-t^2+2=2-t^2$

$K=t^2-2t-t^2+2=2-2t$

$\frac{(2-K)}{2}=t \Rightarrow t^2=2-h$

$(2-K)^2=4(2-h)$

$h \Rightarrow x \quad \& \quad y \Rightarrow y$

$(y-2)^2=4(2-x)$

28. $\therefore 4(\sin^3 A+\sin^3 B+\sin^3 C)+12(\sin A\sin B\sin C)=0$

$\therefore 4(\sin^3 A+\sin^3 B+\sin^3 C-3\sin A\sin B\sin C)=0$

$\Rightarrow \sin A+\sin B+\sin C=0 \quad \sin A=\sin B=\sin C$

29. $\lim_{n\to 0}\frac{(e^{2002x}+1)-(1+1001x)e^{-1001x}-(1+1001x)(e^{1001x}+1)}{x^2\left(\frac{e^x-1}{x}\right)}$

$$\lim_{n\to 0}\left(e^{2002x}+1\right)e^{-1001x}\left[\frac{e^{1001x}-(1+1001x)}{n^2}\right]\left[\frac{x}{e^x-1}\right]$$

$$\lim_{n\to 0} 2\times 1\times\frac{(1001)^2}{2}\times 1=(1001)^2$$

30. let $\log_2^x = t$

$$\left(t^2-t+1\right)\log_2^x=\log_2^{2^{-1+\frac{1}{8}}}$$

$$\left(t^2-t+1\right)t=-\frac{7}{8}$$

$$\Rightarrow t^3-t^2+t+\frac{7}{8}=0$$

$$\Rightarrow\left(t+\frac{1}{2}\right)\left(t^2-\frac{3}{2}+t\frac{7}{4}\right)\qquad D<0 \text{ so imaginary value of } t$$

so, $t=-\frac{1}{2}$

$$\log_2^x=-\frac{1}{2}$$

$$x=2^{-\frac{1}{2}}=(\text{real value})$$

so we get one real value of x.